

MATH 189 HW 1

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Question 1

- a. Regression, $n = 500$, $p = 3$
- b. Categorical, $n = 20$, $p = 14$
- c. Regression, $n = 52$, $p = 3$

Question 2

- a. Obs. 1: 2
Obs. 2: 2
Obs. 3: $\sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.16$
Obs. 4: $\sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.23$
Obs. 5: $\sqrt{(-1)^2 + 1^2} = \sqrt{2} \approx 1.41$
Obs. 6: $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \approx 1.73$
- b. Our prediction for $K = 1$ is Green, because that's the color of the closest neighbor.
- c. Our prediction for $K = 3$ is Red, because it is close to two red neighbors and one green neighbors, so by plurality, red is the color of our prediction.
- d. If the Bayes decision boundary is highly nonlinear, then a low K value would provide the closest KNN decision boundary to Bayes. This is because low K values offer higher flexibility in the KNN decision boundary, which better fits a highly nonlinear Bayes decision boundary.

Question 3

- b. 506 rows and 13 columns. Rows represent each data point, columns represent the entries for each data point.
- c.